

Basic A-Frame VR-Scene

Tool Introduction

Three.js

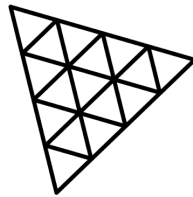


Fig. 8: The Three.js logo.¹

Website: <https://threejs.org/>

HTML just describes structure. To make anything move, react, or exist in three dimensions, you need a separate browser native programming language: JavaScript.

Three.js is a JavaScript library that makes it practical to build and display 3D graphics in the browser. It wraps WebGL, the browser's low-level 3D graphics API, into much simpler to use building blocks: a scene, a camera, lights, and 3D objects, all created and controlled with plain JavaScript code.

Who's behind it: Three.js was created by Ricardo Cabello (known as "Mr.doob") and first released on GitHub in April 2010.² It has since been maintained as a community-driven, open-source project that is not owned or controlled by any single company.³

Remember: Three.js is a large, reusable toolbox of JavaScript code that saves you from writing raw WebGL by hand.

¹Wikimedia Commons, Three.js Icon.svg (2020).

²Wikipedia, Three.js (n.d.).

³GitHub, mrdoob/three.js repository (n.d.).

A-Frame

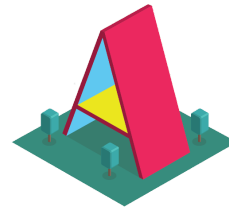


Fig. 9: The A-Frame logo.⁴

Website: <https://aframe.io/>

A-Frame is a free, open-source framework⁵ built on top of [Three.js](https://threejs.org/). It allows you to create VR and WebXR scenes using simple, declarative HTML-like tags (<a-scene>, <a-box>, <a-sky>) instead of writing JavaScript directly.⁶

Who's behind it: A-Frame was originally conceived within Mozilla's VR team and first released in December 2015.⁷ It's now maintained by its original co-creators through Supermedium, a small San Francisco startup founded by three former members of Mozilla's WebVR team.⁸

Remember: Everything you write in A-Frame ultimately becomes Three.js running underneath. A-Frame just lets you describe a 3D/VR scene in HTML-like markup instead of hand-writing that Three.js code yourself.

⁴Inspired To Educate (2017).

⁵GitHub, aframevr/aframe repository (n.d.).

⁶A-Frame, Introduction (aframe.io/docs) (n.d.).

⁷A-Frame, Introduction (aframe.io/docs) (n.d.).

⁸Road to VR, Former Mozilla WebVR Pioneers Launch 'Supermedium' VR Browser (2018).

One Time Per Project Set Up

1. Create a new private repository and clone it

You already know this from the previous chapter.

2. Set up Vite

Vite is a local development server for web projects. We're using it here because it also makes it simple to add HTTPS to that local server, which WebXR requires for testing on a headset.

Open the Terminal in VS

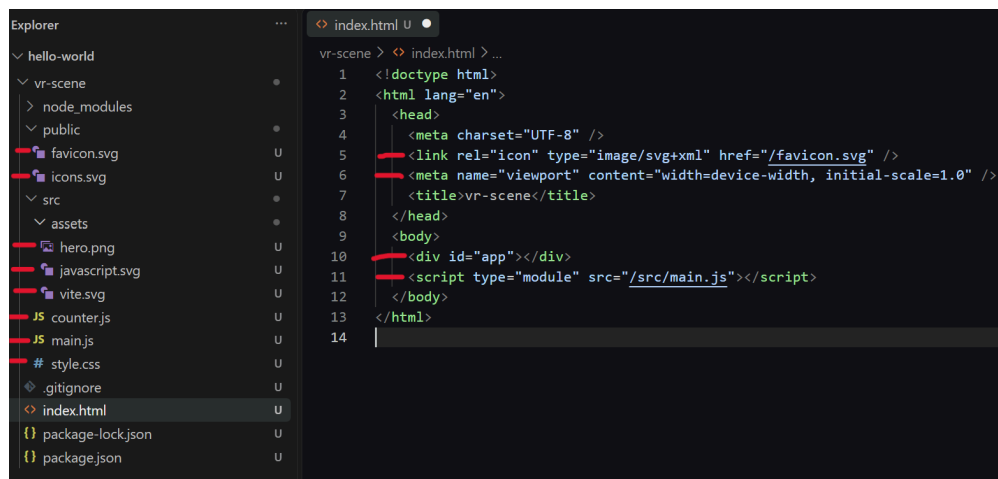
Inside the cloned folder, run `npm create vite@latest` in the VS terminal

You will be guided through the setup process: pick Vanilla / JavaScript when prompted.

3. Delete vite's demo content

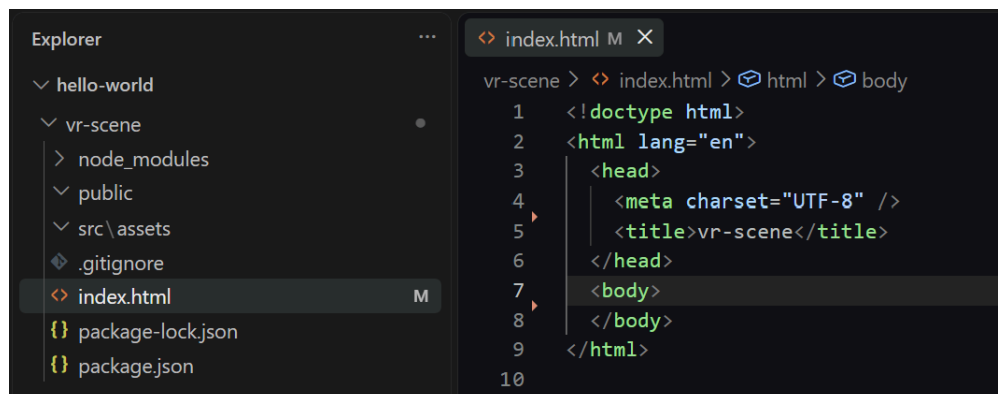
Keep it as a clean, close-to-empty page. Commit and push.

Before



```
1 <!doctype html>
2 <html lang="en">
3   <head>
4     <meta charset="UTF-8" />
5     <link rel="icon" type="image/svg+xml" href="/favicon.svg" />
6     <meta name="viewport" content="width=device-width, initial-scale=1.0" />
7     <title>vr-scene</title>
8   </head>
9   <body>
10    <div id="app"></div>
11    <script type="module" src="/src/main.js"></script>
12  </body>
13 </html>
14
```

After



```
1 <!doctype html>
2 <html lang="en">
3   <head>
4     <meta charset="UTF-8" />
5     <title>vr-scene</title>
6   </head>
7   <body>
8   </body>
9 </html>
10
```

4. Add HTTPS support

WebXR (and camera access for AR) only works in a secure context. That means the page has to be served as HTTPS, not plain HTTP. Your dev server currently only serves plain HTTP, which is fine in a desktop browser but won't work once you open the page on a phone or headset. This one-time setup gives your dev server a (self-signed) HTTPS address that's also reachable from any device on the same Wi-Fi network.

In the terminal Install the plugin with: `npm install --save-dev @vitejs/plugin-basic-ssl9`

Next, configure Vite to use the plugin. Create a `vite.config.js` file in your project's root folder and add:

```
import { defineConfig } from 'vite'
import basicSsl from '@vitejs/plugin-basic-ssl'

export default defineConfig({
  plugins: [basicSsl()],
  server: { host: true },
})
```

4. Run the Vite dev server

- Add again the `<h1>hello world</h1>` in your `index.html` body.
- then, in the terminal, type `npm run dev`

Vite should then give you a local host *localhost*: and a https address that will serve the page in your browser. If you open this https address on your phone or headset (connected to the same wifi) you should see your *hello world*.

Note: When opening the http address for the first time, you'll first see a browser warning like "Your connection is not private". That's expected: this is a self-signed certificate, not one issued by a trusted authority. Click "Advanced" and proceed anyway.

Tip: this only works if you run `npm run dev` in the same folder as your `index.html`. to change the folder, type `cd [your folder containing the index.html]` in your terminal to move to that folder before starting the dev server.

⁹Vite, vite-plugin-basic-ssl repository (n.d.).

Create the Scene

1. Add this basic A-Frame scene.

In index.html, load A-Frame from its CDN and paste in the example scene from A-Frame's own Introduction page:¹⁰ [Introduction – A-Frame](#)

```
<html>
  <head>
    <script
src="https://aframe.io/releases/1.8.0/aframe.min.js"></script>
    </head>
    <body>
      <a-scene>
        <a-box position="-1 0.5 -3" rotation="0 45 0"
color="#4CC3D9"></a-box>
        <a-sphere position="0 1.25 -5" radius="1.25"
color="#EF2D5E"></a-sphere>
        <a-cylinder position="1 0.75 -3" radius="0.5" height="1.5"
color="#FFC65D"></a-cylinder>
        <a-plane position="0 0 -4" rotation="-90 0 0" width="4"
height="4" color="#7BC8A4"></a-plane>
        <a-sky color="#ECECEC"></a-sky>
      </a-scene>
    </body>
  </html>
```

Remember: A-Frame handles the 3D setup, VR entry button, and default camera controls automatically.

7. Test it on your headset

Open the https address Vite gave you on your headset. If the scene doesn't show up, stop the current server (*ctrl + c* in the terminal) and type *npm run dev* again. If everything works, commit and push.

Exercise

Use the A-Frame documentation (no Ai, no imported assets) to:

1. Change the color of one object
2. Add another object (only use A-Frame geometries, not your own 3D objects)
3. Add an animation to an object
4. Experiment!

Remember to commit and push your changes.

¹⁰A-Frame, Introduction (aframe.io/docs) (n.d.).

Who owns the ground we're building on?

Vite

Vite went through exactly the kind of ownership change relevant to this section. It was created by Evan You (also the creator of Vue.js), who founded a company called VoidZero around it in 2024.¹¹ In June 2026, Cloudflare acquired VoidZero. This means that the open source project Vite, a tool millions of projects depend on, is now backed by one of the largest infrastructure companies in the world. Cloudflare explicitly states that Vite stays MIT-licensed and open source. Additionally, it remains "vendor-agnostic" meaning your project isn't locked into Cloudflare's services just because you used Vite. They further state that Vite will keep being developed in the open by the existing Vite team, and Cloudflare is funding a \$1 million independent fund for contributors who aren't Cloudflare or VoidZero employees.¹²

Remember: An acquisition doesn't automatically change how you can depend on a tool. What actually matters is what happens to the license and governance. In this case, both explicitly didn't change.

Three.js and A-Frame

For Three.js is MIT-licensed and maintained by a broad community (100k+ GitHub stars), not dependent on Cabello personally.¹³ For A-Frame, it's thinner: also MIT-licensed (anyone could fork it), but its active maintenance concentrates in one small company (Supermedium) rather than a large community. This is closer to MindAR's situation than Three.js's.¹⁴

Remember: MIT license is your safety net in both cases. Should the worst case ever happen (a maintainer disappearing), you and anyone else are legally free to take the code and keep going yourselves.

¹¹VoidZero, About (voidzero.dev) (n.d.).

¹²Cloudflare, VoidZero Joins Cloudflare (2026).

¹³GitHub, mrdoob/three.js repository (n.d.).

¹⁴A-Frame, Introduction (aframe.io/docs) (n.d.).

References

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